Smart Food Delivery & Marketplace Management System
IE 6700 - Data Management for Analytics

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Problem Statement: Modern food delivery platforms need efficient coordination of multiple stakeholders including customers, restaurants, drivers, and payment systems

Market Context: As cities become busier and more people rely on online platforms for meals, efficient data management becomes critical for business success

Objective: Build well-structured database for order management from browsing to delivery completion

Key Goals:

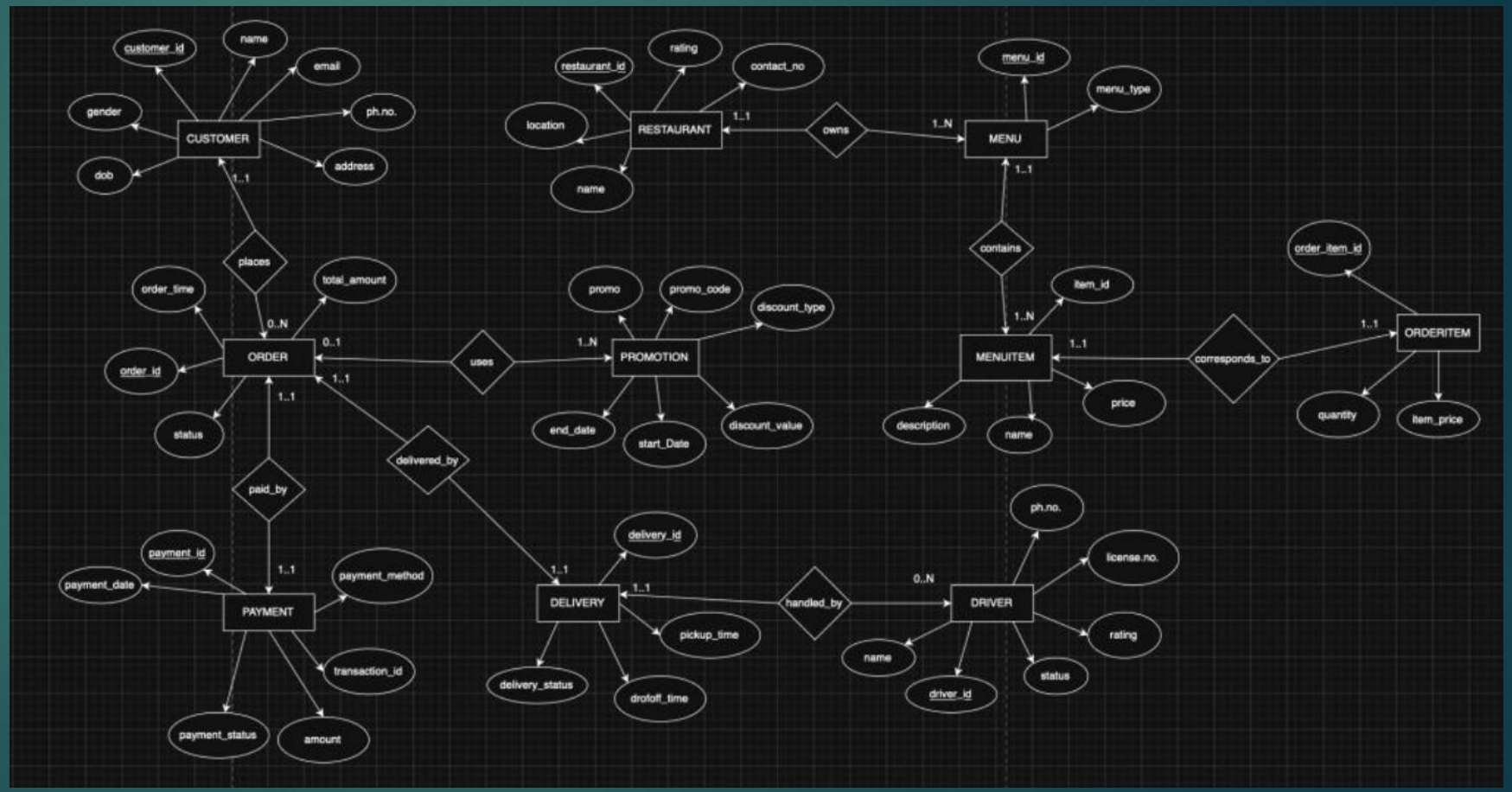
- Organized data storage with proper relationships
- Smooth workflow enabling real-time operations
- Effective decision-making support through analytics

EER Model overview:

Purpose: Visual representation of complete data structure

Key Business Rules:

- Customer (1) → (M) Order
- Menu (1) → (M) Menu_Item
- Order (1) → (1) Payment
- Driver (1) → (M) Delivery
- Driver (1) → (0,1) Vehicle
- Promotion (1) → (M) Order
- Shows relationships:
  1:1, 1:N, N:M
- Displays participation constraints



Database entities:

1. Customer Entity

- PK: Customer_id
- Attributes: name, gender, phone, address, account_type
- Relationship: 1:M with ORDER

2. Restaurant Entity

- PK: Restaurant_id
- Offers multiple menu items
- Relationship: 1:M with Menu

3. Menu Entity

- PK: Menu_id
- FK: Restaurant_id
- Each menu belongs to one restaurant

4. Menu_Item Entity

- PK: Item_id
- Attributes: price, category, availability
- FK: Menu_id

5. Order Entity

- PK: Order_id
- FK: Customer_id, Restaurant_id
- Stores: date, amount, status

6. Order_Item (Bridge Table)

- PK: Order_item_id
- FK: Order_id, Item_id
- Resolves M:N between Order and Menu_Item

7. Payment Entity

- PK: Payment_id
- FK: Order_id
- 1:1 relationship with Order

8. Driver Entity

- PK: Driver_id
- Can deliver many orders

9. Delivery Entity

- PK: Delivery_id
- FK: Order_id, Driver_id
- Tracks: assignments, time, performance

Summary Box:

- 9 Total Entities
- All in 3NF
- Complete system coverage

UML Model Introduction:

UML vs EER:

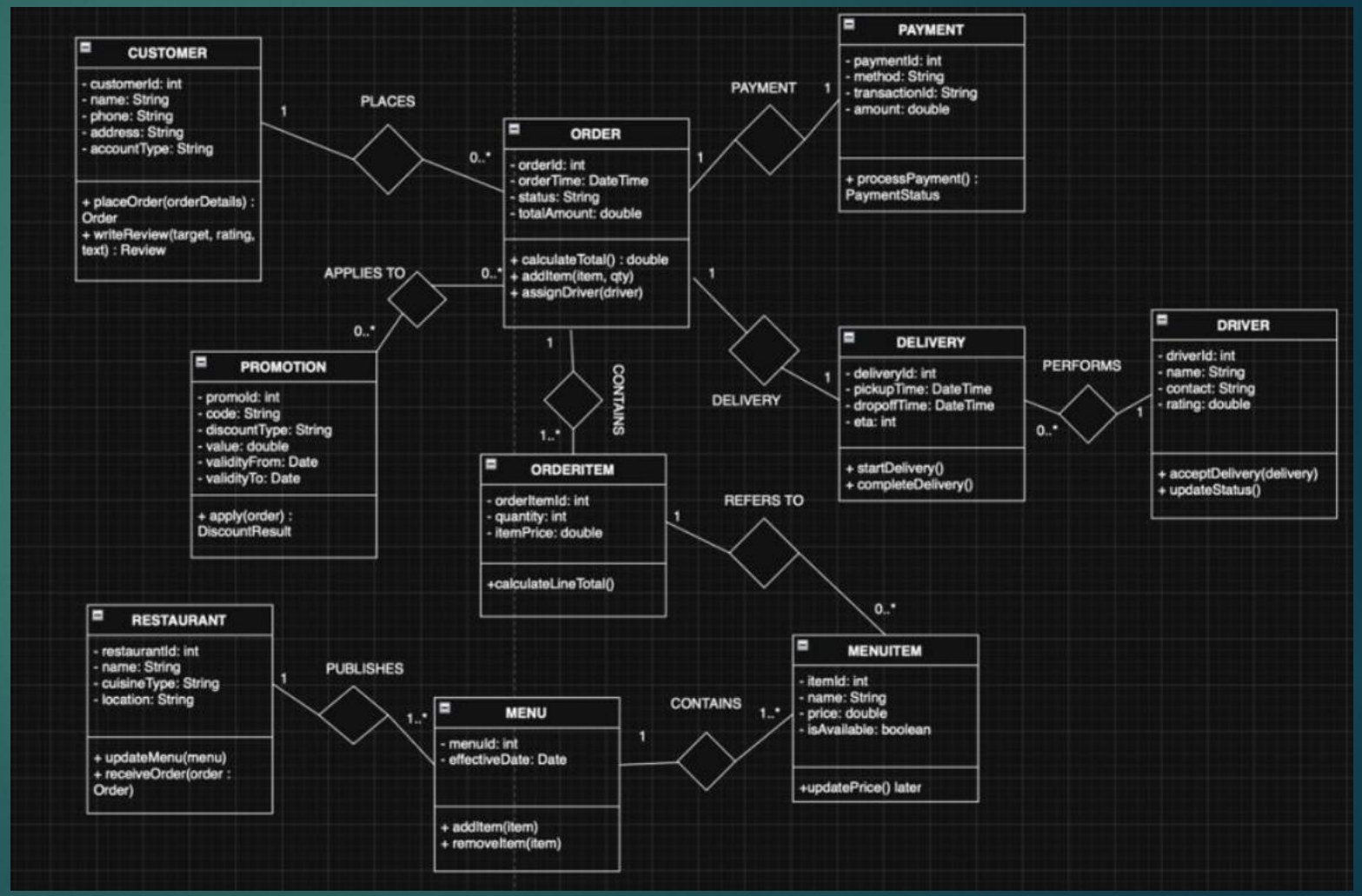
- **EER**: Data storage focus
- **UML**: System behavior focus

Each Class Contains:

- **Attributes**: name, price, rating, status
- **Methods**: placeOrder(), calculateTotal(), processPayment()

Key Relationships:

- Customer *places* Orders
- Orders *contain* OrderItems
- Payments *belong to* Orders
- Deliveries *performed by* Drivers
- Menus *publish* MenuItems
- Promotions *apply to* Orders



Normalization:

All Tables are in 3NF .

Requirements Met:

- ✓ **1NF**: Atomic attribute values, no repeating groups
- ✓ **2NF**: No partial dependencies on composite keys
- ✓ **3NF**: No transitive dependencies

Benefits:

- Data integrity maintained
- No redundancy
- Efficient query performance

Specific Examples:

- CUSTOMER → 3NF (all attributes depend on Customer_id)
- ORDER → 3NF (foreign keys don't violate normalization)
- ORDER_ITEM → 3NF (bridge entity with no redundancy)
- PAYMENT → 3NF (1:1 with Order, no redundancy)

Summary:

Project Deliverables :

- **EER Model** with all 9 entities and 15+ relationships clearly defined and documented
- **UML Model** showing system behavior, methods, and class interactions for implementation guidance
- **Fully Normalized Database** (3NF) ensuring data integrity, no redundancy, and optimal performance
- **Complete System Workflow** from customer browsing through delivery completion and review submission
- **Business Rules Implementation** captured in entity relationships and constraints